The Third Spectral Sum of Graphs: Exact Projection Equivalence, Pole–Pentagon Geometry, and Certified Carrier Reduction

Tony Newton
Newton Astro Labs*
London, UK
tony.newton79@gmail.com

Abstract

Let G be a finite simple graph of order n with adjacency eigenvalues $\lambda_1(G) \geq \cdots \geq \lambda_n(G)$, and write $S_3(G) = \lambda_1(G) + \lambda_2(G) + \lambda_3(G)$ and $M_3(n) = \max_{|V(G)|=n} S_3(G)$. The first result of this paper is an all-rank finite identity: for every $1 \leq k \leq n$, maximizing the Ky Fan k-sum over simple graphs is exactly equivalent to maximizing the entrywise positive part of a rank-k orthogonal projection. Equivalently, if $\eta_{\mathrm{mix}}(k;n)$ denotes the finite mixed projection functional, then $M_k(n) = \frac{n}{2}\eta_{\mathrm{mix}}(k;n) - k$. For $k = 3$ we derive an explicit closed-wheel (pole–pentagon) variational family and prove that its unique interior optimum has coefficient

$$\alpha_3 = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9} = 1.2815681968721678767\ldots,$$

with exact orbit masses $p^* = (91 - 25\sqrt{7})/126$ and $r^* = (7 + 5\sqrt{7})/126$. This gives the unconditional asymptotic lower bound $\liminf_n M_3(n)/n \geq \alpha_3$ and $\eta_{\mathrm{mix}}(3) \geq 2\alpha_3$. A compact computer-assisted certificate bank, when assumed complete, gives the matching upper bound $M_3(n) \leq \alpha_3 n - 3$, the exact value $\eta_{\mathrm{mix}}(3) = 2\alpha_3$, and wheel rigidity for $n \geq 1742$; the remaining replay dependencies are stated explicitly rather than suppressed. Beyond the coefficient calculation, we record self-contained second-order reductions: a Douglas-type range/contraction criterion, an exact two-gauge wedge-area test, an induced-P_3 spanning theorem for the normalization tangent, global elimination of the minimal $2+2$ Perron-silent carrier, and compression of the remaining rank-two signed frame problem to the irreducible $(2,3)$ and $(3,3)$ sectors. These reductions sharply delimit what is proved, what is certificate-dependent, and what remains open in a complete finite classification.

Keywords. adjacency eigenvalues; Ky Fan sums; spectral graph theory; orthogonal projections; projection constants; Parseval frames; interval certification; semidefinite inequalities.
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1 Introduction and prior literature

For a graph G on n vertices let

$$\Lambda_k(G) = \sum_{i=1}^{k} \lambda_i(G), \qquad M_k(n) = \max_{|V(G)|=n} \Lambda_k(G).$$

Partial adjacency spectral sums interpolate between spectral-radius problems, Ky Fan variational geometry, and graph-energy questions [1–4]. The graph constraint matters: after Ky Fan's principle introduces a projection, the 0–1 off-diagonal entries of an adjacency matrix impose a sign-threshold operation absent from a general symmetric-matrix inequality.

1.1 From the general k-sum bound to the third spectral sum

Mohar proved the general estimate

$$M_k(n) \leq \frac{1+\sqrt{k}}{2}\, n,$$

up to the exact finite formulation given in his paper [2]. At $k = 3$ this gives the coefficient $(1+\sqrt{3})/2 = 1.3660254\ldots$. A direct treatment of the third spectral sum already appeared in Ebrahimi Boroojeni's 2008 master's thesis, where the asymptotic coefficient was improved to

$$\tau_3 := \limsup_{n\to\infty} \frac{M_3(n)}{n} \leq \frac{3}{\sqrt{5}} = 1.3416407\ldots.$$

This direct bound is due to Ebrahimi Boroojeni [5]. That thesis belongs to the same line of work as the two-eigenvalue paper of Ebrahimi, Mohar, Nikiforov, and Sheikh Ahmady [6].

The second spectral sum has since been resolved much more sharply. Kumar, Liu, Monterde, Pragada, and Tait proved the conjectured coefficient $8/7$ using graph limits, convex geometry, exterior algebra, and convex optimization [7]. Huang and Wei then determined the exact maximum for every $n \geq 5$ and the equality graph [8]. These developments make $k = 3$ the next natural spectral-sum case rather than merely another instance of a general matrix bound.

*Newton Astro Labs is the trading name under which Tony Newton conducts independent computational research in the United Kingdom; it is not a limited company.

1.2 Projection constants and the 2026 comparison

Projection constants originate in Banach-space geometry [9–13]. Sivashankar, Tang, and Wakhare recently connected graph eigenvalue bounds to absolute projection constants and, in their discussion of sums of largest eigenvalues, isolated the joint or mixed projection functional [14]. Their argument implies

$$\Lambda_k(G) \le \frac{1+\gamma(k)}{2}\, n - k,$$

where $\gamma(k)$ is the maximal absolute projection constant of rank k. For $k = 3$ the classical value $\gamma(3) = (1+\sqrt{5})/2$ gives the coefficient

$$\frac{1+\gamma(3)}{2} = \frac{3+\sqrt{5}}{4} = 1.3090169943\ldots.$$

The three-dimensional projection-constant value is classical [15–17]. The individual third-eigenvalue problem is different from the present $\lambda_1 + \lambda_2 + \lambda_3$ problem; recent work on individual λ_3 and closed blow-ups should therefore be regarded as neighboring, not interchangeable, prior art [18–21].

1.3 Contributions and claim boundary

The paper has two logically distinct layers. The first is analytic and finite: an exact all-rank graph/projection identity, the pole–pentagon lower construction, and several second-order carrier reductions. The second is a computer-assisted compact upper-bound programme. Some legacy source objects for that programme have not yet been restored to a completely independent referee-replay package. Accordingly, all conclusions that use the global upper certificate are stated under an explicit certificate hypothesis. In particular, this paper does *not* claim a complete all-order equality or uniqueness classification.

The principal new structural conclusions recorded here are:

(i) an exact all-k finite identity linking $M_k(n)$ to a mixed projection functional;
(ii) the exact one-variable pole–pentagon optimization giving α_3;
(iii) self-contained range/gain, wedge-area, and induced-P_3 second-order reductions;
(iv) a global exclusion of the minimal $2+2$ Perron-silent carrier; and
(v) a dimension-three coefficient compression reducing the remaining rank-two signed frame frontier to $(2,3)$ and $(3,3)$.

A targeted literature search located the mixed-constant inequality of [14] but not the exact finite all-k equality in theorem 2.1. We therefore state the equality as a mathematical result while keeping the priority claim deliberately narrow.

2 Exact graph–projection equivalence

2.1 The all-rank threshold identity

Let $\mathcal{P}_k(n)$ denote the rank-k orthogonal projections in $\mathbb{R}^{n\times n}$. For a matrix $P = (p_{ij})$, write $P_+ = (\max\{p_{ij}, 0\})_{ij}$ for its entrywise positive part.

Theorem 2.1 (Exact all-rank graph–projection identity)**.** *For every* $1 \le k \le n$,

$$M_k(n) = \max_{P\in\mathcal{P}_k(n)} \big(\mathbf{1}^T P_+ \mathbf{1} - k\big). \tag{1}$$

Proof. Ky Fan's maximum principle gives

$$\Lambda_k(A) = \max_{P\in\mathcal{P}_k(n)} \operatorname{tr}(AP)$$

for every real symmetric A [1, 22, 23]. Fix $P = (p_{ij})$. For a simple adjacency matrix $A = (a_{ij})$,

$$\operatorname{tr}(AP) = 2 \sum_{1\le i<j\le n} a_{ij} p_{ij}.$$

The binary variables $a_{ij} \in \{0, 1\}$ are independent. Their maximizing choice is $a_{ij} = 1$ precisely when $p_{ij} > 0$ (ties at zero are irrelevant), so

$$\max_A \operatorname{tr}(AP) = \sum_{i\ne j} \max\{p_{ij}, 0\}.$$

Because an orthogonal projection is positive semidefinite, $p_{ii} \ge 0$, and $\sum_i p_{ii} = \operatorname{tr} P = k$. Hence the last display equals $\mathbf{1}^T P_+ \mathbf{1} - k$. Maximizing over P proves (1). Conversely, thresholding the off-diagonal signs of any maximizing P produces an honest simple graph, so no relaxation has been introduced. □

2.2 Exact finite mixed projection form

Entrywise, $x_+ = (|x| + x)/2$. Therefore

$$\mathbf{1}^T P_+ \mathbf{1} = \frac{1}{2}\big(\|P\|_1 + \mathbf{1}^T P \mathbf{1}\big), \tag{2}$$

where $\|P\|_1 = \sum_{i,j} |p_{ij}|$ is the entrywise ℓ_1 norm. Define the finite mixed projection functional

$$\eta_{\mathrm{mix}}(k; n) = \frac{1}{n} \max_{P\in\mathcal{P}_k(n)} \big(\|P\|_1 + \mathbf{1}^T P \mathbf{1}\big). \tag{3}$$

Corollary 2.2 (Exact finite graph/mixed-projection equivalence)**.** *For every* $1 \le k \le n$,

$$\boxed{M_k(n) = \frac{n}{2}\, \eta_{\mathrm{mix}}(k; n) - k.} \tag{4}$$

Consequently, if $\eta_{\mathrm{mix}}(k) = \sup_{n\ge k} \eta_{\mathrm{mix}}(k; n)$, *then*

$$\eta_{\mathrm{mix}}(k) = 2 \sup_{n\ge k} \frac{M_k(n) + k}{n}. \tag{5}$$

Remark 2.3 (No Fekete step). Equation (5) is the unconditional asymptotic statement needed here. Disjoint union does not make $M_k(n) + k$ additive or superadditive for fixed k, because the top k eigenvalues of two components are re-ordered after union. No Fekete-type limit argument is used in this paper.

Table 1: Selected upper coefficients relevant to the third spectral sum. The last row is the certificate-dependent candidate upper coefficient of the present work; its lower realization is unconditional.

Source	setting	coefficient for $M_3(n)/n$
Mohar [2]	general k-sum bound at $k=3$	$(1+\sqrt{3})/2 = 1.3660254\ldots$
Ebrahimi Boroojeni [5]	direct asymptotic S_3 analysis	$3/\sqrt{5} = 1.3416407\ldots$
Sivashankar–Tang–Wakhare [14]	mixed/absolute projection bound	$(3+\sqrt{5})/4 = 1.3090169\ldots$
This work	pole–pentagon coefficient	$\alpha_3 = 1.2815681\ldots$

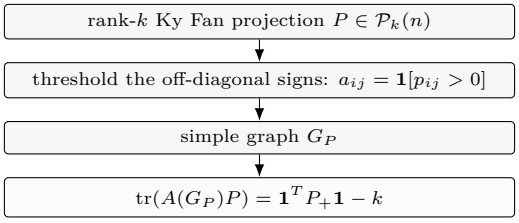


Figure 1: Exact graph–projection threshold mechanism. The graph is recovered from the same Ky Fan projection, so the variational reduction introduces no relaxation.

2.3 Parseval-frame interpretation

Write $P = XX^T$ with $X \in \mathbb{R}^{n\times k}$ and $X^TX = I_k$. If x_i^T is the ith row of X, then

$$p_{ij} = \langle x_i, x_j\rangle, \qquad \sum_{i=1}^{n} x_i x_i^T = I_k.$$

The threshold architecture is summarized in figure 1. Thus theorem 2.1 is an extremal problem over finite Parseval frames [24, 25]: select every positive pairwise inner product, subtract the trace k, and maximize under isotropy. For $k = 3$ this produces a three-dimensional common-carrier problem in which radial mass, angular harmonics, and alignment with $\mathbf{1}$ cannot be optimized independently. Spherical-code and equiangular-line methods provide useful comparison geometry [26–30].

3 Pole–pentagon extremal geometry

3.1 Closed wheel quotient

Let $W_6 = K_1 \vee C_5$ be the six-vertex wheel with one hub and a pentagonal rim. Consider a closed blow-up: each of the six types is replaced by a clique, and two parts are completely joined exactly when their wheel types are adjacent. In normalized weighted form the relevant quotient is

$$H(p,r) = D^{1/2}\big(A(W_6) + I\big)D^{1/2},$$
$$D = \mathrm{diag}(p,r,r,r,r,r).$$

with $p + 5r = 1$ and $0 < r < 1/5$.

The quotient geometry is shown in figure 2. The pentagonal Fourier decomposition splits the quotient into an axial hub/uniform-rim block and two conjugate pairs of rim modes. With $\phi = (1+\sqrt{5})/2$, the two positive non-axial modes both equal $r\phi$. The axial block is

$$\begin{pmatrix} p & \sqrt{5pr} \\ \sqrt{5pr} & 3r \end{pmatrix}. \tag{6}$$

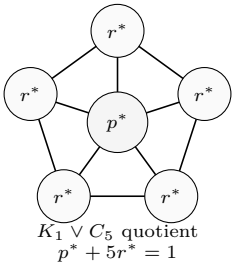


Figure 2: Pole–pentagon geometry of the closed-wheel quotient. In the blow-up construction each displayed type is replaced by a clique of the corresponding asymptotic mass.

Its larger eigenvalue is

$$\lambda_+(r) = \frac{1 - 2r + \sqrt{1 + 4r - 36r^2}}{2}. \tag{7}$$

Therefore the normalized top-three sum of the closed-wheel family is

$$F(r) = \frac{1}{2} + \sqrt{5}\,r + \frac{1}{2}\sqrt{1 + 4r - 36r^2}. \tag{8}$$

3.2 Exact one-variable optimization

Theorem 3.1 (Pole–pentagon variational optimum)**.** *On the feasible interval $0 < r < 1/5$, F is strictly concave and has a unique critical point*

$$r^* = \frac{7 + 5\sqrt{7}}{126}, \qquad p^* = 1 - 5r^* = \frac{91 - 25\sqrt{7}}{126}. \tag{9}$$

At this point

$$F(r^*) = \alpha_3 := \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9}. \tag{10}$$

Proof. Differentiation gives

$$F''(r) = -\frac{20}{(1 + 4r - 36r^2)^{3/2}} < 0$$

throughout the interior feasibility interval. The equation $F'(r) = 0$ reduces, after one exact squaring with the sign fixed by the unsquared stationarity equation, to

$$126r^2 - 14r - 1 = 0.$$

Its positive root is (9). Direct substitution gives $1 + 4r^* - 36(r^*)^2 = 5/7$ and then (10). □

3.3 Unconditional asymptotic lower theorem

Choose integer part sizes whose normalized masses converge to $(p^*, r^*, \dots, r^*)$. The six-type quotient eigenvalues converge continuously to those of $H(p^*, r^*)$, while the within-part closed-blow-up modes contribute only bounded-order corrections. Hence:

Corollary 3.2 (Pole–pentagon lower bound)**.**

$$\liminf_{n\to\infty} \frac{M_3(n)}{n} \geq \alpha_3. \tag{11}$$

Equivalently, by theorem 2.2,

$$\eta_{\mathrm{mix}}(3) \geq 2\alpha_3. \tag{12}$$

The arithmetic comparison with the absolute-projection estimate is exact:

$$\Delta_3 := \frac{3+\sqrt{5}}{4} - \alpha_3 = \frac{9+7\sqrt{5}-4\sqrt{35}}{36} > 0. \tag{13}$$

Notice what (13) does and does not prove. It proves that the *constructed lower coefficient* lies strictly below the decoupled projection upper coefficient. It does not, by itself, prove $\eta_{\mathrm{mix}}(3) < 1+\gamma(3)$; that strict mixed-constant conclusion needs the matching global upper certificate in the next section.

4 Certificate-dependent sharp upper bound

4.1 The compact certificate hypothesis

The compact upper proof developed in the master S3 programme uses exact rational positivity, interval continuation, endpoint congruence checks, Fejer–Riesz factorization, and rational positive-trace duals. To keep the logical status transparent we package only the global implication needed in this paper.

Certificate hypothesis 4.1 (Compact upper certificate C)**.** *All exact compact-region certificate objects used to exclude non-pole–pentagon rank-three projection states are valid and cover their declared domains, including the endpoint, continuation, Bernstein-sign, Fejer–Riesz, and rational-dual components needed for the universal upper estimate.*

The hypothesis is mathematical, not a claim about archive completeness. The legacy replay ledger is discussed separately in section 8.

4.2 Conditional sharp coefficient and mixed constant

Theorem 4.2 (Sharp rank-three coefficient conditional on C)**.** *Assume theorem 4.1. Then every simple graph G of order $n \geq 3$ satisfies*

$$S_3(G) \leq \alpha_3 n - 3, \qquad M_3(n) \leq \alpha_3 n - 3. \tag{14}$$

Together with theorem 3.2, this yields

$$\lim_{n\to\infty} \frac{M_3(n)}{n} = \alpha_3 \tag{15}$$

and

$$\eta_{\mathrm{mix}}(3) = 2\alpha_3 = 1 + \frac{\sqrt{5}}{9} + \frac{2\sqrt{35}}{9}. \tag{16}$$

Proof. The certificate implication gives (14). The lower sequence in theorem 3.2 forces equality of the asymptotic coefficient. Finally (5) converts the same upper/lower pair into (16) without any disjoint-union argument. □

Corollary 4.3 (Conditional rank-three coupling deficit)**.** *Assume theorem 4.1. For every $Q \in \mathcal{P}_3(N)$,*

$$\big(N - \mathbf{1}^T Q \mathbf{1}\big) + \big(\gamma(3)N - \|Q\|_1\big) \geq 2\Delta_3 N, \tag{17}$$

and the coefficient $2\Delta_3$ is asymptotically sharp.

Proof. The two terms on the left are the deficits from the separate bounds $\mathbf{1}^T Q \mathbf{1} \leq N$ and $\|Q\|_1 \leq \gamma(3)N$. Their sum is $(1+\gamma(3))N - (\|Q\|_1 + \mathbf{1}^T Q \mathbf{1})$. Apply (16) and (13). □

4.3 Conditional wheel rigidity

The finite certificate architecture also contains an order-specific entrance analysis into the equality collar and a wrong-chamber continuation.

Theorem 4.4 (Wheel rigidity within the certificate boundary)**.** *Assume theorem 4.1 together with the order-specific equality-collar certificates used in the finite programme. If $n \geq 1742$ and G attains $M_3(n)$, then G is a clique blow-up of $K_1 \vee C_5$.*

The threshold 1742 is therefore a certified rigidity threshold *inside* the universal certificate boundary. It is not used below as an unconditional input, and it does not settle the finite equality problem for $11 \leq n \leq 1741$.

5 Second-order common-carrier architecture

The remaining finite problem is not controlled by first-order eigenvalue counts alone. A useful exact organization comes from the Hessian of the weighted template problem. Let $M(w) = D_{\sqrt{w}} H D_{\sqrt{w}}$, assume $\lambda_3 > \lambda_4$, and let u_a be an orthonormal eigenbasis. For a simplex tangent h put $x = W^{-1}h$, $W = \mathrm{diag}(w)$. The spectral perturbation formula separates the modes below the top-three cluster into positive and negative tail contributions [31–33].

For $a \leq 3 < b$, set $v_{ab} = u_a \circ u_b$. The coefficient

$$-\frac{4\lambda_a\lambda_b}{\lambda_a - \lambda_b}$$

is positive when $\lambda_b < 0$ and negative when $0 < \lambda_b < \lambda_3$. Collecting the corresponding rank-one terms gives positive semidefinite Gram matrices N and P with $R = N - P$.

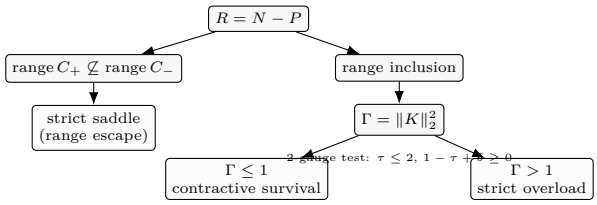


Figure 3: Second-order decision structure behind the Douglas factorization. Range inclusion is necessary but not sufficient; the residual transfer must also be contractive.

5.1 Exact Hessian congruence and Douglas factorization

Theorem 5.1 (Mass-tangent/all-tail congruence)**.** *At a stationary point with $\lambda_3 > \lambda_4$,*

$$D^2 S_3(w)[h,h] = -\frac{1}{2} x^T R x, \qquad x = W^{-1}h, \qquad R = N - P. \tag{18}$$

Hence a positive second variation is exactly a failure of $R \succeq 0$ on the normalization tangent.

The factorization of the survival condition is a finite-dimensional instance of Douglas's range/factorization theorem [34].

Proposition 5.2 (Range/contraction form)**.** *Let $N = C_- C_-^T$ and $P = C_+ C_+^T$. Then*

$$N - P \succeq 0 \iff C_+ = C_- K, \quad \|K\|_2 \le 1 \text{ for some } K. \tag{19}$$

Thus survival requires both range inclusion and contractive gain. Range inclusion alone is not sufficient.

5.2 Two-gauge scalar and wedge-area criteria

The logical role of range escape and gain overload is summarized in figure 3. When a boundary/closure reduction produces a two-dimensional gauge bank, the contraction test is completely scalar. Let $N_G = C_{-,G} C_{-,G}^T \succ 0$ and $P_G = C_{+,G} C_{+,G}^T \succeq 0$, and assume range inclusion. Define

$$\tau = \operatorname{tr}(N_G^{-1} P_G), \qquad \delta = \frac{\det P_G}{\det N_G}. \tag{20}$$

Theorem 5.3 (Two-gauge overload criterion)**.** *If Γ_G is the largest generalized gain of (P_G, N_G), then*

$$\Gamma_G \le 1 \iff \tau \le 2 \ \text{ and } \ 1 - \tau + \delta \ge 0. \tag{21}$$

Proof. The eigenvalues $\mu_1, \mu_2 \ge 0$ of $N_G^{-1/2} P_G N_G^{-1/2}$ satisfy $\mu_1 + \mu_2 = \tau$ and $\mu_1 \mu_2 = \delta$. Since $1 - \tau + \delta = (1 - \mu_1)(1 - \mu_2)$, the two scalar signs are equivalent to $\max\{\mu_1, \mu_2\} \le 1$. □

Let the negative-tail columns be $a_i \in \mathbb{R}^2$ and the positive-tail columns $b_j \in \mathbb{R}^2$. With $u \wedge v = \det[u\ v]$, define

$$A_- = \sum_{i<k} (a_i \wedge a_k)^2,$$

$$A_+ = \sum_{j<\ell} (b_j \wedge b_\ell)^2,$$

$$A_\times = \sum_{i,j} (a_i \wedge b_j)^2.$$

Cauchy–Binet and the two-dimensional Lagrange identity give

$$A_- = \det N_G, \qquad A_+ = \det P_G, \qquad A_\times = \operatorname{tr}(\operatorname{adj}(N_G) P_G). \tag{22}$$

Consequently:

Corollary 5.4 (Wedge-area form)**.** *Under the hypotheses of theorem 5.3,*

$$\Gamma_G \le 1 \iff A_\times \le 2A_- \ \text{ and } \ A_\times \le A_- + A_+. \tag{23}$$

This exact local test is one of the main reductions of the higher-inertia branch. What remains open is a universal topology-to-area implication forcing one of these inequalities to fail (or forcing range escape) for every reduced above-line non-wheel carrier.

5.3 Induced-P_3 tangent frame

Let B be a connected noncomplete negative template. For an oriented induced path $a - b - c$ put

$$p_{abc} = e_a + e_b - 2e_c. \tag{24}$$

Lemma 5.5 (Induced-P_3 spanning)**.**

$$\operatorname{span}\{p_{abc} : a - b - c \text{ induced}\} = \mathbf{1}^\perp. \tag{25}$$

Proof. Every p_{abc} has coordinate sum zero. If y is orthogonal to both orientations of every induced P_3, then the two equations force $y_a = y_b = y_c$ on each such path. If an edge uv had $y_u \ne y_v$, it could lie in no induced P_3, so u and v would be true twins. A shortest path to a nonneighbor, or a nonedge joined through universal twins, then creates induced P_3's forcing $y_u = y_v$, a contradiction. Connectedness therefore makes y constant. □

Combining theorems 5.1 and 5.5 shows that any $n - 1$ independent induced-P_3 tangents form an exact local frame for the all-tail Hessian. A single negative diagonal of the corresponding R-matrix is a strict-saddle certificate.

5.4 Tail-dispersion identity

For $p = p_{abc}$, let q_i denote the top-three normalized row vectors and let $r_{\ell i}$ denote the normalized row of tail mode $\ell > 3$. Define

$$O_p = \|q_a - 2q_c\|^2 + \|q_b\|^2, \qquad A_p = q_a q_a^T + q_b q_b^T - 2q_c q_c^T,$$

$$z_\ell = r_{\ell a} q_a + r_{\ell b} q_b - 2 r_{\ell c} q_c, \qquad \eta_\ell = \lambda_\ell^2 z_\ell^T (\Lambda - \lambda_\ell I)^{-1} z_\ell.$$

Then the exact induced-P_3 dispersion formula is

$$\frac{1}{2} D^2 S_3[p,p] = O_p - \operatorname{tr}(\Lambda A_p^2) + \sum_{\ell > 3} \eta_\ell, \qquad \eta_\ell \ge 0. \tag{26}$$

Thus any finite tail subset can provide a rigorous saddle certificate; a finite diagnostic census is not needed as a logical part of the identity.

6 Global closure of the minimal $2+2$ carrier

The second-order reduction leaves a Perron-silent rank-two transverse carrier in which the positive and negative response Gram matrices agree. The smallest nontrivial support has two noncollinear generators on each sign side. The master analysis reduces its local attachment grammar to one singular seven-type family; the late continuation closes that family globally.

6.1 Exceptional local family

For the canonical seven-type positive template on (a, b, p, q, c, d, x), exact characteristic-polynomial arithmetic gives

$$\chi_P(\lambda) = \lambda(\lambda-1)(\lambda+1)(\lambda^4 - 7\lambda^3 + 13\lambda^2 - \lambda - 8), \quad (27)$$

so a Sturm count yields

$$\text{inertia}(P) = (4, 2, 1),$$
$$\ker P = \text{span}\{(1, 0, -2, -1, 1, 0, 1)^T\}.$$

Positive diagonal congruence preserves this inertia and transports the kernel accordingly.

6.2 Global range saturation and twin forcing

Let $A = DPD$ with D positive diagonal. Because A already uses all four allowed positive directions, an exterior coupling vector r must be orthogonal to $\ker A$: otherwise the compression to the kernel direction and the new exterior coordinate has negative determinant and contributes an additional positive direction. After the positive weights cancel, this is an exact combinatorial linear condition on the exterior incidence word.

Combining it with the zero-level fourth-mode balance leaves only six of the $2^7 = 128$ exterior negative-neighbor words. Their exact Schur complements are

$$0,\ 0,\ 2,\ 0,\ 2,\ 1. \quad (28)$$

The positive values create a fifth positive direction; each zero value is a column already present in the quotient. Every globally legal exterior extension is therefore a zero-Schur twin of an existing type.

6.3 Uniform twin-quotient exclusion

The resulting weighted twin quotient admits a rational positive-trace majorant with exact positive-definiteness checks. Under the stationary weight relations it gives the uniform ceiling

$$S_3 \le \frac{453077}{375000} = 1.208205\ldots < \alpha_3. \quad (29)$$

Theorem 6.1 (Global minimal-carrier closure)**.** *The complete minimal $2+2$ Perron-silent shared-edge carrier is globally excluded from the pole–pentagon extremal coefficient. The conclusion uses only the local singular-family classification, kernel-forced twin extension, and the exact rational ceiling* (29)*.*

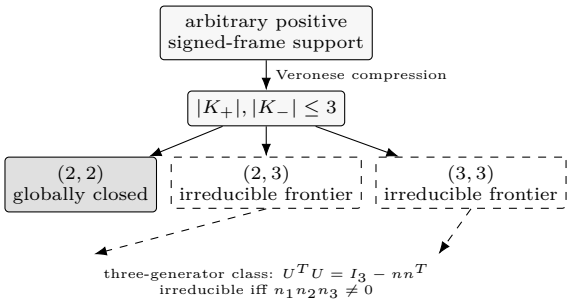


Figure 4: Exact support compression and the surviving Naimark frontier. The continuous frame equality alone reduces the problem to two sectors but does not discretize their incidence geometry.

This theorem is independent of the unresolved $(2,3)$ and $(3,3)$ higher-support sectors. It removes the entire lowest-support obstruction rather than a finite sample of its representatives.

7 Higher-support compression and Naimark frontier

7.1 Constructive compression in Sym_2

Suppose

$$K_+ = \sum_i a_i v_i v_i^T = \sum_j b_j w_j w_j^T = K_-,$$
$$K_+ = K_- \succ 0, \qquad a_i, b_j > 0. \quad (30)$$

The vector space of real symmetric 2×2 matrices has dimension three. Under the Veronese map $(x, y) \mapsto (x^2, xy, y^2)$, any side of (30) with more than three active generators has a nonzero coefficient-kernel direction. Orient such a direction h so some $h_i > 0$ and set

$$t = \min_{h_i > 0} \frac{a_i}{h_i}, \qquad a_i' = a_i - t h_i.$$

Then all $a_i' \ge 0$, at least one coefficient vanishes, and the represented matrix is unchanged. Iteration terminates.

Theorem 7.1 (Signed frame support compression)**.** *Every positive representation* (30) *can be replaced, without changing $K_+ = K_-$, by one satisfying*

$$|K_+| \le 3, \qquad |K_-| \le 3. \quad (31)$$

If K has rank two, each nonzero sign class needs at least two generators. Modulo sign reversal and theorem 6.1, the only irreducible support cardinalities are therefore $(2,3)$ *and* $(3,3)$*.*

7.2 Three-generator Naimark coordinate

Whiten a three-generator sign class so that $U = [u_1\ u_2\ u_3] \in \mathbb{R}^{2\times 3}$ and $UU^T = I_2$. Put

$$\Delta_{ij} = \det(u_i, u_j), \qquad n = (\Delta_{23}, \Delta_{31}, \Delta_{12})^T.$$

The Pluecker relation gives $Un = 0$, while Cauchy–Binet gives $\|n\| = 1$. Therefore

$$U^T U = I_3 - nn^T. \tag{32}$$

If $u_i = (x_i, y_i)^T$, direct determinant expansion gives

$$\det \begin{pmatrix} x_1^2 & x_2^2 & x_3^2 \\ x_1 y_1 & x_2 y_2 & x_3 y_3 \\ y_1^2 & y_2^2 & y_3^2 \end{pmatrix} = -\Delta_{12}\Delta_{23}\Delta_{31} = -n_1 n_2 n_3. \tag{33}$$

Corollary 7.2 (Irreducible three-generator sector)**.** *A whitened three-generator sign class is coefficient-compressible to two exactly when at least one $n_i = 0$. Its irreducible sector is therefore*

$$n_1 n_2 n_3 \neq 0. \tag{34}$$

The complete support-routing picture is summarized in figure 4. This converts the remaining continuous frame layer into a compact Naimark modulus plus the discrete incidence, Perron, and stationarity equations. It also explains why a finite topology list cannot be justified from the frame equality alone.

8 Verification, reproducibility, and conclusion

8.1 Independent algebraic checks

The exact constants in section 3, the stationary quadratic $126r^2 - 14r - 1$, the value $F(r^*) = \alpha_3$, the second derivative of F, and the determinant identity (33) were independently recomputed by symbolic algebra for this release. The small-order graph check in section B exhausts the NetworkX graph atlas through order seven [35–39]. These checks are consistency tests; they are not substituted for analytic proofs or interval certificates.

The computer-assisted parts use standard exact and validated-numerics tools: integer-preserving elimination [40], interval methods [41–44], numerical linear-algebra error control [45, 46], and polynomial/semidefinite positivity methods [47–52]. Fejer–Riesz factorization supplies the trigonometric positivity interface [53, 54].

8.2 Current certificate-release boundary

The analytic graph/projection identities, pole–pentagon optimization, Douglas-type factorization, two-gauge algebra, induced-P_3 spanning result, and higher-support compression are self-contained at the level stated here. The minimal $2 + 2$ structural reduction is likewise analytic through twin forcing, while its final uniform quotient ceiling uses the finite exact rational majorant certified in the Version-62 S3 continuation; this local certificate is distinct from the unresolved legacy compact upper bank. By contrast, the complete compact universal upper proof still has partial replay dependencies: the later all-chart pass transcript has been recovered, but the complete endpoint source/input package for two desingularized charts is not; a rank-$(4, 3)$ report records 7467 accepted continuation leaves, but its raw leaf archive/replayer is not yet restored; the historical Bernstein raw input is partial; and the canonical final rational-dual/Fejer–Riesz source bundle is incomplete. Therefore this release does not describe theorem 4.2 as an independently replay-complete theorem.

The finite classification is also not complete. The minimal $2 + 2$ carrier is closed, but the irreducible $(2, 3)$ and $(3, 3)$ frame sectors still require the incidence/Perron/stationarity constraints. The whole-core interval ledger and the universal topology-to-area implication outside the recovered low-inertia universe likewise remain active obligations. These are mathematical frontiers, not formatting qualifications.

8.3 Prior-art boundary

The paper does not claim the invention of Ky Fan's variational principle, the mixed projection constant, the maximal absolute projection constant, the Douglas factorization theorem, or general Parseval-frame and spherical-code machinery. Those ingredients are classical or recent prior art [1, 9, 14, 24, 34]. The original contribution asserted here is the specific graph-threshold exactification in theorem 2.1, the rank-three pole–pentagon evaluation and its integration with the S3 certificate programme, and the S3-specific carrier reductions in sections 5 to 7. The candidate universal coefficient is not labeled unconditional until the global certificate hypothesis is discharged in a fully replayable release.

8.4 Conclusion

The third spectral sum admits a cleaner structure than a direct graph census suggests. The graph maximization can be performed exactly inside the Ky Fan variational principle, producing the finite identity $M_k(n) = n\eta_{\mathrm{mix}}(k; n)/2 - k$ at every rank. At rank three the closed wheel reduces the candidate extremal geometry to one concave variable and yields the exact pole–pentagon coefficient α_3. The remaining obstruction is not a vague search over all graphs: the all-tail Hessian has a range/gain factorization, two-dimensional closure banks admit exact area inequalities, induced paths span the full normalization tangent, the minimal $2 + 2$ carrier is globally eliminated, and higher signed frame support compresses to the two irreducible sectors $(2, 3)$ and $(3, 3)$.

This is a substantial narrowing of the S3 problem while preserving the line between an unconditional theorem and a certificate-dependent conclusion. A complete release now has a precise target: finish the replay-complete compact upper certificate and close the residual higher-support incidence/stationarity sectors. If those gates are discharged, the same document immediately yields $M_3(n) \leq \alpha_3 n - 3$ and $\eta_{\mathrm{mix}}(3) = 2\alpha_3$ with no change to the unconditional derivations.

A Exact constants and comparison algebra

The constants used throughout satisfy

$$\alpha_3 = 1.28156819687216787675243501399\ldots,$$

$$2\alpha_3 = 2.563136393744335750350487002798\ldots,$$
$$\gamma(3) = \frac{1+\sqrt{5}}{2} = 1.6180339887498948482\ldots,$$
$$\frac{1+\gamma(3)}{2} = 1.3090169943749474241\ldots,$$
$$\Delta_3 = 0.02744879750277954735\ldots,$$
$$1+\gamma(3)-2\alpha_3 = 2\Delta_3.$$

For the pole–pentagon masses,

$$p^* = 0.19727156526496218\ldots,$$
$$r^* = 0.16054568694700756\ldots.$$

and $p^* + 5r^* = 1$ exactly.

B Exhaustive small-order sanity check

An exhaustive traversal of the standard graph atlas through order seven gives

n	$M_3(n)$	comment
3	0	all three eigenvalues sum to zero
4	2	exhaustive
5	$1+\sqrt{5}$	3.2360679775..., exhaustive
6	4.6855577203...	wheel degree sequence $(5,3,3,3,3,3)$
7	5.9376300962...	exhaustive

These values lie below $\alpha_3 n - 3$ at the tested orders. The check is included only as a finite consistency audit; it does not establish the global upper bound.

C Certificate dependency ledger

For publication purposes the global status can be compressed to four rows:

Object	Current status
Analytic S3 reductions	Self-contained in the mathematical argument.
Compact universal upper bank	Mathematical implication retained under C; independent legacy replay still partial.
Finite $2+2$ carrier	Globally closed by exact twin forcing and rational quotient ceiling.
Higher-support finite frontier	Reduced to irreducible $(2,3)$ and $(3,3)$ sectors; complete finite uniqueness withheld.

D Determinant check for the Naimark sector

For $u_i = (x_i, y_i)^T$, expansion of the Veronese determinant gives

$$\det\begin{pmatrix} x_1^2 & x_2^2 & x_3^2 \\ x_1y_1 & x_2y_2 & x_3y_3 \\ y_1^2 & y_2^2 & y_3^2 \end{pmatrix}$$
$$= -(x_1y_2 - x_2y_1)(x_2y_3 - x_3y_2)(x_3y_1 - x_1y_3),$$

which is $-\Delta_{12}\Delta_{23}\Delta_{31}$. This is precisely the criterion used in theorem 7.2: vanishing of one Pluecker coordinate makes the three Veronese columns dependent and permits coefficient compression to two generators.

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